STATS 7022 Data Scence PG Formula Sheet

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1 Linear Regression

Models

Bias of $\hat{f}$ $b_f = \mathrm{E}[\hat{y}_i] - y_i$

Bias-Var Trade-off $\mathrm{E}[(y_0 - \hat{y}_0)^2] = \mathrm{var}(\hat{y}_0) + \mathrm{Bias}(\hat{y}_0)^2 + \mathrm{var}(\epsilon)$

TSS $= \sum_{i=1}^{n} (y_i - \bar{y})^2$ **RSS** $= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

MSE(LOSS) $= \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ **F** $= \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)}$

$\mathbf{R}^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}$ **RMSE** $= \sqrt{\frac{\mathrm{RSS}}{n-p-1}}$

MLR

Linear Model: $Y = X\beta + e$ with $E[\epsilon] = 0$ and $\mathrm{var}(\epsilon) = \sigma^2 I_{n\times n}$

Estimator of β: $Q(\beta) = \|y - X\beta\|^2$

Estimator of variance $(\sigma^2): {s_e}^2 = \frac{1}{n-p-1} \|y - X\hat{\beta}\|^2$

Slope: $\hat{\beta} = (X^T X)^{-1} X^T y$

Simple LR: $\hat{\beta}_1 = \sum_{i=1}^{n} \frac{x_i - \bar{x}}{\sum_{i=1}^{n} (x_i - \bar{x})^2} y_i \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

2 Classification

ROC

Sensitivity(Recall) $= \frac{TP}{TP+FN}$

Specificity $= \frac{TN}{TN+FP}$

Precision $= \frac{TP}{TP+FP}$

FPR $= \frac{FP}{FP+TN}$

TPR $= \frac{TP}{TP+FN}$

F1-Score $= \frac{2TP}{2TP+FP+FN}$

Box-Cox Transformation

$$y = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(x), & \lambda = 0 \end{cases}$$

Naive Bayes

$$P(Y=k|X=x) = \frac{1}{Z} P(Y=k) \prod_{i=1}^{p} P(X_i = x_i | Y=k)$$

LDA & QDA

LDA: $\delta_k(x) = \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$

LDA (p > 1): $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$

QDA: $\delta_k(x) = -\frac{1}{2} x^T \Sigma_k^{-1} x + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \log|\Sigma_k| + \log(\pi_k)$

3 Model Selection

Subset Selection

Given model M_0 for null model, M_p for full model, and M_k for k predictors,

- **Best Subset Selection:** For $k = 1, 2, ..., p$, fit all $\binom{p}{k}$ models and choose the best one.
- **Forward Stepwise Selection:** Start with M_0 and add one variable at a time.
- **Backwards Stepwise Selection:** Start with M_p and remove one variable at a time.

Ridge and Lasso

Ridge Regression

$$min\{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \hat{\beta}_j^2\}$$

Lasso Regression

$$min\{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda |\hat{\beta}_j|\}$$

$RSS(\lambda) = (y - X\hat{\beta})^T (y - X\hat{\beta}) + \lambda \beta^T \beta$, solving gives $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$

Metrics

$$C_p = \frac{1}{n}(RSS + 2p\hat{\sigma}^2)$$

$$C_p' = \frac{RSS}{\hat{\sigma}^2} + 2p - n$$

$$AIC = 2p - 2log(\hat{L})$$

$$AICc = AIC + \frac{2p(p+1)}{n-p-1}$$

$$BIC = log(n)p - 2log(\hat{L}).$$

Adjusted $R^2 = 1 - \frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)}$

LOOCV

$$\mathrm{MSE}_i = (y_i - \hat{y}_i)^2 \quad CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} MSE_i,$$

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2, \quad h_i = diag(X(X^T X)^{-1} X^T)$$

4 Polynomial Regression and Spline

Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + ... + \beta_d x_i^d + \epsilon_i$$

Giving K knots, we have K+1 intervals.

$$C_0(X) + C_1(X) + ... + C_{K-1}(X) + C_K(X) = 1.$$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + ... + \beta_K C_K(x_i) + \epsilon_i$$

Cubic Spline

A cubic spline with K knots,

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + ... + \beta_K C_K(x_i) + \epsilon_i.$$

Truncated polynomial basis functions:

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x-\xi)^3, & x > \xi \\ 0, & x \leq \xi \end{cases}$$

- Freedom of Cubic Spline: $K+4$ degrees of freedom.
- A degree-d spline is a piecewise degree-d polynomial with continuity in derivatives up to degree d $-$ 1.

- To fit a cubic spline with K knots, we fit a linear regression with an intercept and the K + 3 predictors: $x, x^2, x^3, h_1(x,\xi_1), h_2(x,\xi_2), ..., h_K(x,\xi_K)$.
- A natural cubic spline with K knots uses K degrees of freedom

Smoothing Splines

We can try and ensure that g(x) is smooth by choosing g(x) that minimizes $\sum_{i=1}^{n}(y_i - g(x_i))^2 + \lambda \int g''(x)^2 dx$.

Effective Degrees of Freedom: $df_\lambda = \sum_{i=1}^{n}[S_\lambda]_{ii}$, the trace of S_λ.

5 KNN

KNN Regression

$$\hat{P}(Y=j|X=x_0) = \frac{1}{K}\sum_{i\in\mathcal{N}_0} I(y_i = j)$$

$$\hat{f}(x_0) = \frac{1}{K}\sum_{i\in\mathcal{N}_0} y_i$$

Nadaraya–WatsonKernel-WeightedAverage

$$\hat{f}(x_0) = \frac{\sum_{i=1}^{n} k_\lambda(x_0, x_i) y_i}{\sum_{i=1}^{n} k_\lambda(x_0, x_i)}$$

Epanechnikov quadratic kernel

$$K_\lambda(x_0, x) = D\left(\frac{|x - x_0|}{\lambda}\right)$$

$$D(t) = \begin{cases} \frac{3}{4}(1-t^2), & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

6 MARS and CART

MARS

$$\hat{f}(x) = \sum_{i=1}^{k} c_i B_i(x)$$

where $B_i(x)$ takes one of three forms:

- an intercept,
- a hinge function $max(0, x-c)$ or $max(0, c-x)$
- a product of hinge functions $h_j(x)h_k(x)$ for some ξ_j and ξ_k.

GCV

$$\text{GCV} = \frac{\frac{1}{N}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\left(1 - \frac{ENP}{N}\right)^2}$$

where $ENP = k + \frac{d(k-1)}{2}$ and penalty term d is usually 2 or 3.

Michaelis-Menten Model

$$f(x, (K, V_m)) = \frac{V_m x}{K + x}$$

Classification Trees

$$\hat{p}_{mk} = \frac{1}{N_m}\sum_{i\in R_m} I(y_i = k), \quad k(m) = \underset{k}{\text{argmax}}\, \hat{p}_{mk}$$

Misclassification rate

$$\frac{1}{N_m}\sum_{x_i\in R_m} I(y_i \neq k(m)) = 1 - \hat{p}_{mk(m)}$$

Gini Index

$$\sum_{k=1}^{K}\hat{p}_{mk}(1-\hat{p}_{mk}) = 1 - \sum_{k=1}^{K}\hat{p}_{mk}^2$$

Cross entropy / deviance

$$-\sum_{k=1}^{K}\hat{p}_{mk}\log(\hat{p}_{mk})$$

7 Random Forest

Bootstrap

Consider a random sample $S = \{X_1, X_2, ..., X_n\}$, and its bootstrap resample:

$$S_b^* = \{X_{b1}^*, X_{b2}^*, ..., X_{bn}^*\},$$

where

$$\bar{X^*} = \hat{\text{E}}[X^*] = \frac{1}{B}\sum_{b=1}^{B} X_b^* \qquad \hat{\text{var}}(X^*) = \frac{1}{B-1}\sum_{b=1}^{B}(X_b^* - \bar{X^*})^2 \qquad \bar{b}_T(\theta)(\textbf{Bias}) = \hat{X}^* - X$$

Bagging

Consider $\{X_1, X_2, ..., X_n\}$, $X_i \sim N(\mu.\sigma^2)$. Then,

$$\text{var}(\bar{X}) = \frac{\sigma^2}{n} \qquad \text{var}(\bar{X}) = \rho\sigma^2 + \frac{1-\rho}{n}\sigma^2 \quad \text{with correlation } \rho$$

Guidelines

- **Classification:** Default for m is $\lfloor\sqrt{p}\rfloor$ and minimum node size of 1
- **Regression:** Default for m is $\lfloor\frac{p}{3}\rfloor$ and minimum node size of 1

SVM

P-dimensional Hyperplane

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p = 0$$

Properties

$$yi(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p) > 0$$

Inner Product

$$\langle x_i, x_{i'}\rangle = x_i^T x_{i'} = \sum_{j=1}^{p} x_{ij}x_{i'j}$$

Kernel functions

Linear

$$k(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij}x_{i'j}$$

Polynomial

$$k(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij}x_{i'j}\right)^d$$

Radial

$$k(x_i, x_{i'}) = \exp\left(-\gamma\sum_{j=1}^{p}(x_{ij} - x_{i'j})^2\right)$$

8 PCA

Consider p predictors $X_1, X_2, ..., X_p$,

First PC

$$Z_1 = \Phi_{11}X_1 + \Phi_{21}X_2 + \cdots + \Phi_{p1}X_p, \quad \sum_{j=1}^{p} {\Phi_{p1}}^2 = 1$$

$$Z_1 = {\alpha_1}^T X$$

$$var({\alpha_1}^T X) = {\alpha_1}^T \sum \alpha_1$$

Lagarange

$${\alpha_1}^T \sum \alpha_1 - \lambda({\alpha_1}^T \sum \alpha_1 - 1)$$

Differentiating w.r.t. α_1 and setting to zero, $\Rightarrow \sum \alpha_1 = \lambda \alpha_1$

$$\Rightarrow \text{Max}{\alpha_1}^T \sum \alpha_1 = \text{Max}{\alpha_1}^T \lambda \alpha_1 = \text{Max}\lambda {\alpha_1}^T \alpha_1 = \text{Max}\lambda$$

Second PC

$$\text{Max}{\alpha_2}^T \sum \alpha_2, \quad \text{uncorrelated with } {\alpha_1}^T X.$$

$$cov({\alpha_1}^T X, {\alpha_2}^T X) = {\alpha_1}^T \sum \alpha_2 = {\alpha_2}^T \sum \alpha_1 = {\alpha_2}^T \lambda \alpha_1 = \lambda {\alpha_2}^T \alpha_1$$

Lagarange

$${\alpha_2}^T \sum \alpha_2 - \lambda({\alpha_2}^T \alpha_2 - 1) - \Phi({\alpha_2}^T \alpha_2 - 0)$$

Differentiating w.r.t. α_2 and setting to zero,

$$\Rightarrow \sum \alpha_2 - \lambda \alpha_2 - \varphi \alpha_1 = 0.$$

$$\Rightarrow {\alpha_1}^T \sum \alpha_2 - \lambda {\alpha_1}^T \alpha^2 - \Phi {\alpha_1}^T \alpha_1 = 0,$$

$$\Rightarrow \Phi = 0$$

$$\Rightarrow \sum \alpha_2 = \lambda \alpha_2.$$

9 K-means Clustering

Notation

Let $C_1, ..., C_k$ denote sets containing indices of each observation in each cluster.

- $C_1 \cup C_2 \cup \cdots \cup C_k = \{1, 2, ..., n\}$
- $C_k \cap {C_K}' = \varnothing$ for all $k \neq k'$.

Clustering

$$\min_{C_1, C_2, ..., C_k} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \right\}$$

Note that

$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \bar{x}_{kj})^2,$$

where

$$\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}.$$

Algorithm

1. Randomly assign a number of 1 to K to each observation.
2. Iterate until no changes in cluster assignment:
   (a) For each cluster, calculate the centroid. The centroid is the vector of the sample mean of the p features for observations in the cluster.
   (b) Assign each observation to the cluster of the centroid that is closest.

10 Multi-dimensional Scaling

Scaling Procedures

- Classical scaling: dissimilarities are Euclidean distances.
- Ordinal scaling: uses only the order of the dissimilarity measure.

Dissimilarity Coefficients

The dissimilarity coefficient between two points s and r must satisfy the following constraints:

1. $d_{sr} \geq 0$ for every r and s.
2. $d_{rr} = 0$ for every r.
3. $d_{sr} = d_{rs}$ for every r and s.
4. $d_{rt} + d_{ts} \geq d_{rs}$ for every r, s and t (Only if A metric satisfies 1-3).

Dissimilarity Coefficients

Euclidean

$$d_{rs} = \left(\sum_{i=1}^{p} (x_{ri} - x_{si})^2 \right)^{\frac{1}{2}}$$

Manhattan

$$d_{rs} = \sum_{i=1}^{p} |x_{ri} - x_{si}|$$

Minkowski

$$d_{rs} = \left(\sum_{i=1}^{p} |x_{ri} - x_{si}|^R \right)^{\frac{1}{R}}$$

Association Table

	Subject r	
Subject s	1	0
1	a	b
0	c	d

Simple matching coefficient

$$\frac{a+d}{a+b+c+d}$$

Hamming distance

$$b+c$$

Jaccard's coefficient

$$\frac{a}{a+b+c}$$

Classical Scaling

Consider

$$B = XX^T$$

The (r,s)th term of B

$$b_{rs} = \sum_{j=1}^{p} x_{rj} x s j.$$

Euclidean Distances

$$d_{rs}^2 = \sum_{j=1}^{p} (x_{rj} - x_{sj})^2 = b_{rr} + b_{ss} - 2b_{rs}.$$

Summing Over r, s, and Both

$$\sum_{r=1}^{n} d_{rs}^2 = T + n b_{ss}$$

$$\sum_{s=1}^{n} d_{rs}^2 = n b_{rr} + T$$

$$\sum_{r=1}^{n} \sum_{s=1}^{n} d_{rs}^2 = 2nT$$

where

$$T = \sum_{r=1}^{n} b_{rr},$$

i.e., the trace of B.

Rearranging gives

$$b_{rs} = -\frac{1}{2} \left[d_{rs}^2 - d_{r\bullet}^2 - d_{s\bullet}^2 + d_{\bullet\bullet}^2 \right].$$

Example Using Two Normals

Consider a model with two normally distributed groups. For ease of explanation assume that each group has the same variance.

$$Y_1 \sim N(\mu_1,\sigma^2),$$
$$Y_1 \sim N(\mu_1,\sigma^2), \text{ and}$$
$$Y = (1-\triangle)Y_1 + \triangle Y_2,$$

where $\triangle in\{0,1\}$ and

$$P(\triangle = 1) = \pi.$$

Mixture Density

$$f(y;\pi,\mu_1,\mu_2,\sigma^2) = (1-\pi)f_1(y;\mu_1,\sigma^2) + \pi(y;\mu_2,\sigma^2)$$

where f_1 and f_2 are the respective PDFs of the normal distributions.

Log-likelihood

$$\ell(\theta;y) = \sum_{i=1}^{n} log((1-\pi)f_1(y_i;\mu_1,\sigma^2) + \pi f_2(y_i;\mu_2,\sigma^2))$$

Latent Variable

Let Δ_i be latent variables such that if $\Delta_i = 1$ then observation $i \sim \mathcal{N}(\mu_2,\sigma^2)$. With this missing information, the likelihood becomes easier to deal with:

$$f(y_i;\pi,\mu_1,\mu_2,\sigma^2,\Delta_i) = \left[(1-\pi)f_1(y;\mu_1,\sigma^2)\right]^{1-\Delta_i}\left[\pi f_2(y;\mu_2,\sigma^2)\right]^{\Delta_i}.$$

Latent Variable Log-likelihood

$$\begin{aligned}\ell(\theta;\mathbf{y},\Delta) &= \sum_{i=1}^{n}(1-\Delta_i)\log\big((1-\pi)f_1(y_i;\mu_1,\sigma^2)\big) + \Delta_i\log\big(\pi f_2(y_i;\mu_2,\sigma^2)\big)\\ &= \sum_{i=1}^{n}(1-\Delta_i)\log(1-\pi) + (1-\Delta_i)\log\big(f_1(y_i;\mu_1,\sigma^2)\big)\\ &\quad + \sum_{i=1}^{n}\Delta_i\log(\pi) + \Delta_i\log\big(f_2(y_i;\mu_2,\sigma^2)\big)\end{aligned}$$

Responsibilities

Since we don't know Δ_i, we use the expected value

$$\gamma_i(\theta) = \mathbb{E}[\Delta_i \mid \theta,\mathbf{y}] = \mathbb{P}[\Delta_i = 1 \mid \theta,\mathbf{y}].$$

EM Algorithm for Two-component Normal Distribution

1. Make initial guesses of parameters: $\hat{\pi},\hat{\mu}_1,\hat{\mu}_2,\hat{\sigma}_1^2,\hat{\sigma}_2^2$.

2. *Expectation step:* compute the responsibilities:

$$\hat{\gamma}_i = \frac{\hat{\pi} f(y_i;\hat{\mu}_2,\hat{\sigma}^2)}{(1-\hat{\pi})f(y_i;\hat{\mu}_1,\hat{\sigma}^2) + \hat{\pi} f(y_i;\hat{\mu}_2,\hat{\sigma}^2)}$$

3. *Maximisation step:* estimate the parameters:

$$\hat{\mu}_1 = \frac{\sum_{i=1}^{n}(1-\hat{\gamma}_i)y_i}{\sum_{i=1}^{n}(1-\hat{\gamma}_i)}, \qquad \hat{\mu}_2 = \frac{\sum_{i=1}^{n}\hat{\gamma}_i y_i}{\sum_{i=1}^{n}\hat{\gamma}_i},$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(1-\hat{\gamma}_i)(y_i-\hat{\mu}_1)^2 + \hat{\gamma}_i(y_i-\hat{\mu}_2)^2}{n-2},$$

$$\text{and} \qquad \hat{\pi} = \frac{\sum_{i=1}^{n}\hat{\gamma}_i}{n}.$$

4. Iterate steps 2 and 3 until convergence.